



Turbidity Structures the Controls of Ecosystem Metabolism and Associated Metabolic Process Domains Along a 75-km Segment of a Semiarid Stream

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ABSTRACT

Stream ecosystem metabolism contributes to global carbon cycling, yet predicting metabolism across ecosystems remains elusive. Even within stream segments, spatial variation in metabolic rates and their controls can be substantial, exhibiting sudden rather than continuous changes. We measured metabolism at 6 sites along a 75-km mainstem segment of Marsh Creek, Idaho. This agricultural stream lacks major geomorphic transitions such as tributaries or changes in valley width, but possesses patchy patterns of turbidity. We asked: (1) How variable is metabolism along this segment, and (2) How do the controls on metabolism vary along this segment? Metabolism varied several-fold along this stream segment. Average rates of gross primary

production (GPP), ecosystem respiration (ER), and net ecosystem production (NEP) among sites did not correlate with water quality or aquatic macrophyte cover. Rather, reaches along this segment appear to represent different process domains that were characterized by turbidity: More turbid reaches saw negative effects of turbidity on GPP and NEP and positive effects on ER. Turbidity was associated with increased respiration and decoupled ER from GPP at turbid sites. Less turbid sites had stronger coupling between GPP and ER, and GPP was predicted by light and temperature. Heterogeneous sediment supply and transport capacity along Marsh Creek create patchy patterns of turbidity, which affects the controls on metabolism. Geomorphic controls on ecosystem processes are complex but not random. Understanding how metabolism and its controls vary across process domains is needed to scale up process rates and understand how process rates might respond to future changes.

Key words: Ecosystem metabolism; Process domains; Reach scale; Turbidity; Geomorphology; Stream.

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HIGHLIGHTS

- Process domains along stream segment affect controls on ecosystem metabolism.
- In turbid reaches, turbidity decreased GPP and increased ER.
- Total suspended solids stimulated ER, and decoupled GPP and ER at turbid sites.

INTRODUCTION

Lotic ecosystems are responsible for a quarter ($0.32 \text{ Pg C year}^{-1}$) of total CO_2 fluxes from all inland waters (Battin and others 2008; Drake and others 2017). Rates of stream metabolism influence overall carbon cycling by mineralizing as much as half of terrestrially derived carbon (Cole and others 2007), and thus contribute significantly to global CO_2 fluxes (Battin and others 2009; Hotchkiss and others 2015; Drake and others 2017). Most of what is known about how stream metabolism varies over time and space comes from studies of individual reaches that are used to investigate variation across stream order (for example, Vannote and others 1980; Minshall and others 1983; Ulseth and others 2018), environmental gradients within a region (for example, Izagirre and others 2008; Yates and others 2013; Saltarelli and others 2018), or among regions (Lamberti and Steinman 1997; Mulholland and others 2001; Bernot and others 2010; Dodds and others 2015). However, metabolic rates and even regimes—the seasonal patterns of GPP and ER—can vary within watersheds (Mejia and others 2018). Due to the extreme spatial and temporal variability in metabolism within and among streams, predicting patterns of metabolism across ecosystems, and even within stream segments, has been elusive (Hall and others 2015; Bernhardt and others 2017; Siders and others 2017; Dodds and others 2018).

One challenge in predicting metabolism is that first-order controls on metabolism are not consistent predictors of metabolism among sites (Tank and others 2010). As a result, the controls of metabolism at any given site are difficult to predict and can vary across space, time, and scale (Roberts and others 2007; Bernot and others 2010; Hall and others 2015). Differences in correlates with metabolism among studies may result from the range of conditions included within a study (for example, variation in GPP over space may not be related to light among sites with similar light conditions). However, differences in relationships between

metabolism and environmental factors are also likely due to interactions among environmental controls. For example, light may only drive GPP under conditions of low turbidity (Hall and others 2015) or where nutrient concentrations are sufficient to support growth (Myrstener and others 2021).

Predicting ecosystem metabolism across systems could be aided if metabolism and its controls vary systematically across structural gradients such as geomorphology. A variety of stream ecology concepts focus on how geomorphological processes affect habitat and disturbance regimes and therefore ecosystems structure and processes (Swanson and others 1988; Montgomery 1999; Poole 2002; Thorp and others 2006). Studies that have evaluated these concepts in the field have compared ecosystems with major differences in geomorphology such as confined and unconfined valleys (Bellmore and Baxter 2014), stream planform (Venarsky and others 2018), or average watershed slopes (Jankowski and others 2014; Jankowski and Schindler 2019). Yet, metabolism varies within stream and river systems that lack these major geomorphic transitions (Hall and others 2015; Siders and others 2017; Dodds and others 2018; Rocher-Ros and others 2019).

The controls on ecosystem metabolism, including geomorphological controls, are likely more complex than current conceptual models would indicate. For example, turbidity often strongly decreases GPP (Izagirre and others 2008; Griffiths and others 2013; Hosen and others 2014; Hall and others 2015) and, as a proxy for suspended sediment, is controlled by geomorphic processes such as hillslope erosion, bank erosion, and riverine storage and remobilization (Walling 1983; de Vente and others 2007). Yet, variation in suspended sediment or turbidity is not always correlated with geomorphic features (Webster and Cadenasso 2020), and may result from organic inputs or discrete events. Process domains, defined as areas within which distinct geomorphic processes and associated temporal dynamics in disturbances control ecosystem structure and function (Montgomery 1999), may be less systematic or predictable than originally envisioned. There is substantial variation in ecosystem function even within systems that may appear homogeneous (for example, Dodds and others 2018), but how that variation is controlled, either by geomorphic processes or interactions between geomorphic and ecological phenomena, is yet unclear.

We assessed how rates of and controls on stream metabolism varied along a 75-km segment of an

agricultural stream. This segment has no major geomorphic transitions that would traditionally be considered delimiters of process domains (for example, perennial tributaries, dams, major changes in valley constriction; Bellmore and Baxter 2014; Wohl 2018), but varies in channel slope (Figure 1) and sinuosity. A combination of natural geomorphology (that is, subtle changes in channel slope, Figure 1) and human management (channelization, agricultural impacts, and attempts to mitigate these impacts) results in irregular patterns of turbidity along this segment (Guilinger 2017; Meese 2018). We hypothesized that these spatial patterns in turbidity would set a template for heterogeneous spatial patterns of metabolic process rates and their controls. We asked: (1) How variable is metabolism along a 75-km stream segment? And (2) How do the controls on metabolism vary along this segment?

METHODS

Study Area

We studied six reaches along a 75-km stream segment of Marsh Creek, a semiarid agricultural stream located in southeastern Idaho (Figure 1; Table 1). The studied segment of Marsh Creek begins from an outlet of a shallow pond and continues as a low-gradient stream ultimately draining approximately 1100 km² of irrigated agriculture, dryland farming, rangelands, and steep, forested mountains before joining the Portneuf River in Inkom, ID. Local precipitation averages approximately 45 cm annually, and high flow events are

due to snow melt and rain-on-snow in late winter and/or early spring. Average discharge was 3.1 m³/s at the outlet and ranged from 0.9 to 20.3 m³/s during the study period, October 2016 through mid-October 2017. Historically, Marsh Creek has had elevated concentrations of suspended sediments, phosphorus, nitrogen, and *E. coli*, resulting in it being listed as “impaired” under section 303(d) of the Clean Water Act (McSorley 1977; IDEQ 2010; IASCD 2013). There are no perennial tributaries along the study segment, and agricultural land use intensifies patterns of low flow, turbidity, algal and macrophyte biomass, and nutrient concentrations.

Geomorphically, this segment of Marsh Creek has no major transitions that are typically associated with process domains such as perennial tributaries or major changes in valley constriction (Poole 2002; Benda and others 2004; Thorp and others 2006). Yet there is variation in channel planform and gradient with potential ecological importance. Sinuosity ranges from 1 (linear) to nearly 2.5 (sinuous), largely dictated by anthropogenic straightening since the 1940s (Meese 2018). Straightening shortened the stream length, steepening the channel, increasing stream power, and enabling erosion of fine grained banks (Meese 2018). The steepened channel loses elevation in discrete steps, often associated with small agricultural check dams (Figure 1B). Slope varies dramatically along the stream, ranging from 0 to 0.005 (mean 0.0008 ± 0.0009 standard deviation, Figure 1C). Each low-gradient, slack water pool upstream of a step functions as a transient reservoir.

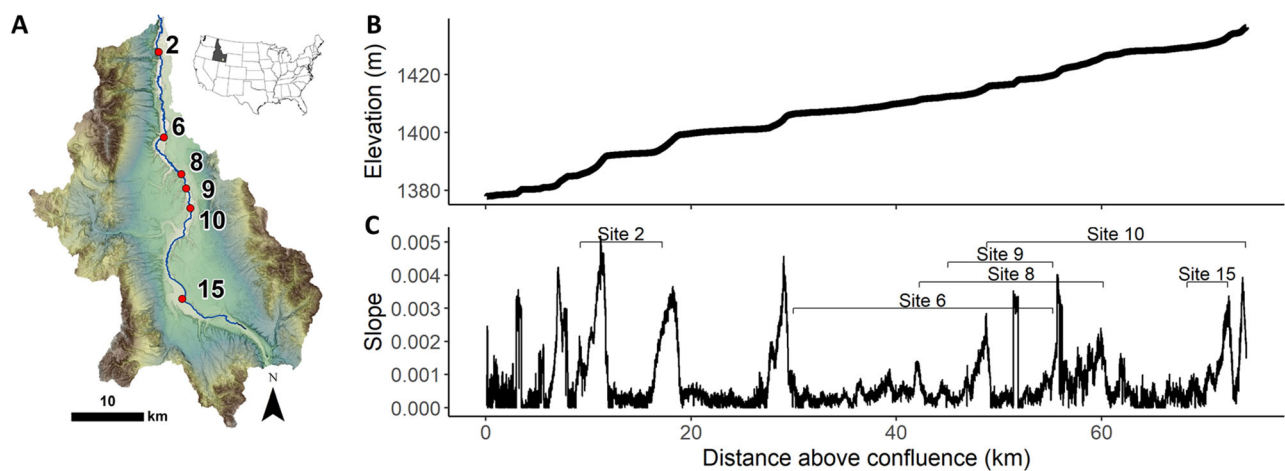


Figure 1. **A** Marsh Creek watershed with study sites indicated in red. Longitudinal patterns of **B** water surface elevation (m above sea level) and **C** slope (unitless) along the studied segment. Parentheses in subpanel **C** indicate mean integrated reach extent for each site (50% turnover). Elevation and slope derivatives were derived from a 1-m airborne lidar digital elevation model.

Table 1. Site Locations, Dates of Sonde Deployments, Total Number of Days with Successful Metabolism Models, and Mean (Standard Deviation) of Growing Season Metabolism Estimates

Site	Distance upstream from confluence (km)	Dominant substrate	Date of deployment	Total days of modeled metabolism (N)	Growing season days of modeled metabolism (N)	Mean (SD) growing season GPP ($\text{g O}_2 \text{ m}^{-2} \text{ d}^{-1}$)	Mean (SD) growing season ER ($\text{g O}_2 \text{ m}^{-2} \text{ d}^{-1}$)	Mean (SD) growing season NEP ($\text{g O}_2 \text{ m}^{-2} \text{ d}^{-1}$)	% Growing season days autotrophic
2	9.2	Gravel	April 19, 2017	129	123	1.38 (0.80)	- 1.68 (0.53)	- 0.29 (0.79)	46
6	29.9	Gravel	May 25, 2017	126	126	1.67 (0.69)	- 1.45 (0.63)	0.23 (0.34)	79
8	42.3	Fine sediment	April 19, 2017	113	100	2.25 (1.05)	- 2.38 (0.96)	- 0.14 (0.31)	32
9	45	Fine sediment	October 1, 2016	276	120	4.15 (1.95)	- 4.78 (2.05)	- 0.63 (0.97)	27
10	48.8	Gravel	October 1, 2016	255	140	1.55 (0.87)	- 2.31 (1.09)	- 0.76 (0.54)	1
15	68.3	Fine sediment	October 1, 2016	189	106	4.51 (3.60)	- 5.55 (2.85)	- 1.03 (1.56)	25

During low flow, suspended sediment settles from the water column, decreasing downstream turbidity. During high flow events, impounded fine sediment is remobilized and flushed downstream (Guilinger 2017). We expected this variation in geomorphology and turbidity to affect patterns of stream metabolism along the stream. Although historic sediment inputs to Marsh Creek were largely attributed to upland erosion (Meese 2018), contemporary sources of suspended sediment are primarily sourced from eroding mainstem banks (Guilinger 2017). Turbidity is highest in the upstream section of the segment (Site 15), with a general decline in turbidity over the next ~20 river km, likely due to reduced sediment conveyance (Guilinger 2017). Turbidity then increases in the lower 30 km of Marsh Creek, consistent with increases in slope and discharge which amplify stream power and, consequently, bank erosion (Guilinger 2017; Meese 2018).

Sensor Measurements

We estimated whole-stream metabolism at the reach scale using the one-station, open water method (Odum 1956; Hall 2016) using field-calibrated YSI 6920 V2 multiparameter sondes (Yellow Springs Instrumentation, Yellow Springs, OH, USA). Sonde locations (Figure 1) were determined by landowner permissions. Each sonde recorded dissolved oxygen, turbidity, specific conductance, and water temperature at 15-min intervals, except at Site 6 where a turbidity sensor failed. During the year-long deployment from October 2016 through mid-October 2017, sondes were calibrated in the field using water-saturated air approximately every 2 weeks. Turbidity and specific conductance probes were calibrated in the field when standard calibration criteria were not met.

Light intensity (LUX) was recorded every 15 min with HOBO temperature and light intensity loggers (UA-002-64; Onset, Bourne, MA, USA) mounted above the stream surface adjacent to the sondes. Paired LUX and photosynthetically active radiation (PAR) readings from a HOBO PAR sensor over a 30-day period in the summer were used to create a linear relationship ($\text{PAR} = 0.00847 * \text{LUX}$, $R^2 = 0.94$, $p < 0.001$, Long and others 2012) that was used to convert all light intensity measurements to PAR for the study.

Barometric pressure was measured near Site 2 using a pressure transducer (HOB0 U20-001-04 Water Level Logger; Onset, Bourne, MA, USA) until March 2017. After this date, we used hourly barometric pressure data for the Pocatello airport

weather station obtained from the NOAA National Climate Data Center.

Water level was measured at 15-min intervals using pressure transducers (HOB0 U20-001-04 Water Level Logger; Onset, Bourne, MA, USA) placed adjacent to each sonde. Manual discharge measurements using either a handheld Acoustic Doppler Velocimeter (FlowTracker; SonTek, San Diego, CA, USA) during wadeable conditions or an Acoustic Doppler Current Profiler (M9, SonTek, San Diego, CA, USA) during higher-flow conditions were used to create stage–discharge relationships for each reach (Guilinger 2017). We did not make direct measures of water level at all study sites after March 2017 because water level loggers were removed from all sites except Site 6, which is a USGS gaging station. We used discharge at Site 6, as well as dummy variables for snowmelt and the seasonal flow recession, to estimate discharge at the other sites. Models were developed for each site using data from March to October 2016 and were used to estimate discharge at all sites from February 1 to October 13, 2017. Predicted discharge was compared to measured flow data for February 2017 at each site to quantify model error. These estimates of error are likely larger for February than the March–October period because flows are higher during this month than the estimated flow period. R^2 for these relationships ranged from 0.73 to 0.98 (see Supplemental materials for more details on discharge estimates). Discharge–velocity relationships were determined from manual discharge measurements at each site and were used to estimate mean velocity for each day during the study period.

Much of Marsh Creek is either channelized or naturally incised, simplifying channel geometry to a rectangular prism with consistent wetted width. We used 1-m LiDAR to measure 20 channel widths over a 200-m reach above each sonde location. Daily reach-averaged stream depth was calculated as $\text{Discharge}/(\text{Velocity} \times \text{Reach-Averaged Wetted Width})$. The assumption of a rectangular channel morphology was reasonable for most sites but may lead to overestimates of depths at Sites 6 and 8 (see Supplemental materials for channel cross sections).

To derive the longitudinal profile and channel slope (Figure 1), elevation was extracted every 5 m along the channel centerline of Marsh Creek from a 1-m airborne LiDAR digital elevation model. Channel slope, width, and stream power were averaged over the mean 50% turnover length (see below for turnover length calculations) for each site.

Periodic Sampling

Water samples were collected monthly at each site along Marsh Creek (Figure 1) and analyzed for total phosphorus (TP), total dissolved nitrogen (TDN), nitrate (NO_3^-), ammonium (NH_4^+), soluble reactive phosphorus (SRP), and dissolved organic carbon (DOC). Water samples for dissolved solutes were filtered through ashed Whatman GF/F filters within 24 h of collection and frozen until analysis. TP samples were frozen immediately until analysis. Samples for NO_3^- , NH_4^+ , and SRP were analyzed on a Lachat QuikChem FIA system (Hach, Loveland, CO, USA) at the Environmental Analytical Laboratory at Brigham Young University (detection limit = 0.02 mg/L, precision within 10%). Samples for TDN and DOC were analyzed on a Shimadzu TOC/TN analyzer also at Brigham Young University (detection limit = 0.07 mg/L, precision within 5% and detection limit = 0.2 mg/L, precision within 10% respectively). TP samples were digested using a persulfate digestion protocol and analyzed as SRP as above.

Sestonic chlorophyll *a* was measured monthly concurrent with water samples from May 2017 to September 2017. Following an extraction in 90% methanol, samples were centrifuged, and the supernatant was analyzed for chlorophyll *a* concentration using a spectrophotometer at the Idaho State University's Center for Ecological Research and Education. Total suspended solids (TSS) were measured as the total dry mass captured on a pre-weighed ashed GF/F filter. Samples were then ashed at 400 °C for four hours to quantify ash-free dry mass (AFDM) in the water column (Steinman and others 2007).

Instream growth of macroalgae and macrophytes was estimated as percent cover for 11 transects every 10 m for 100 m upstream of the sonde using the Braun-Blanquet method (Bowden and others 2007). Surveys were done monthly during the growing season from May 2017 to September 2017.

Metabolism Modeling

We used inverse modeling to estimate GPP, ER, and K_{600} from observed dissolved oxygen time series using “streamMetabolizer” package in R (Appling and others 2018a, 2018b). Within streamMetabolizer, we used the model formulation: `b_Kb_oipi_tr_plrckm.stan`. This is a Bayesian hierarchical state space model that includes both process and observation errors. The model estimates K_{600} using partial pooling based on mean daily discharge; this approach minimized challenges

associated with equifinality and inaccurate parameter estimates. Briefly, this approach leverages the relatively long time series and pools K_{600} toward a piecewise relationship between K_{600} and log daily discharge for each site (see Appling and others 2018a for details). Without pooling K_{600} estimates, ER and K_{600} estimates were significantly correlated; using the partial pooling approach resolved this problem. In our final dataset, ER and K_{600} were not correlated within any sites (see Supplemental materials Figure S4 for plots of ER- K_{600} relationships for each site).

Days with unrealistic estimates of GPP and ER (negative GPP and positive ER) were removed from the final dataset ($N = 80$ days with negative GPP and 17 days with positive ER) and were not used in any subsequent analyses. We also removed 67 days with unrealistically low estimates of K for a given discharge. Some sites had many more instances of unrealistic K estimates that affected site standard deviations (for example, the coefficient of variation for K /discharge among sites ranged from 0.3 to 3), so we were unable to apply a consistent cutoff (for example, using a z -score of > 2) across sites. Instead, we identified outlier days by visually inspecting time series of K , discharge, and discharge/ K . This resulted in a total of 1085 valid days among all sites (Table 1).

To evaluate the spatial independence of metabolism estimates among sites, we estimated 50% and 80% turnover lengths of O_2 using modeled K_{600} and velocity calculated from discharge-velocity relationships. Using a 50% turnover length ($0.7K_{\text{O}_2}/V$; Demars and others 2015), there were 648 instances of non-independence between adjacent sites (60% of modeled days). Using the more conservative 80% turnover length ($1.61K_{\text{O}_2}/V$; Hall and others 2015), there were 892 instances of non-independence between adjacent sites, or 82% of our modeled days. Using an 80% turnover length, Site 2 overlapped with Site 6, but not with any upstream sites. Sites 6, 8, 9, and 10 overlapped for nearly all modeled days. Site 15 was most independent from Site 9 (38% overlapping days; Table 2). Overlap varied among adjacent site pairs, with overlap on most days for Sites 6, 8, and 9, but less frequent overlap with adjacent upstream sites for Sites 2 and 10 (Figure S5). Average 50% turnover length for each site is illustrated by parentheses in Figure 1c.

To assess how sensitive our metabolism estimates were to our reach-averaged depth estimates, we also calculated rates of GPP and ER using the channel-averaged depth at each sonde. These channel-averaged depth estimates were derived

Table 2. Percent of Modeled Days in Which 50% and 80% Turnover Length Overlapped Between Pairs of Sites

Site	% Modeled days overlapping with Site (50%/80% turnover)				
	6	8	9	10	15
2	0%/35%	0%/1%	0%/0%	0%/0%	0%/0%
6		87%/98%	74%/98%	61%/95%	13%/68%
8			100%/100%	98%/100%	13%/82%
9				59%/99%	10%/38%
10					30%/59%

from the depth–discharge relationships obtained from manual discharge measurements. Estimates of average depth at the sonde was not more likely to be accurate than our estimates using LiDAR-estimated wetted widths. However, the different shapes of the depth–Q relationships from these two methods provided a point of comparison and allowed us to test how our findings might be sensitive to our depth estimation method (Table S1).

Analyses

Question 1: Variation in Whole-Stream Ecosystem Metabolism

We analyzed variation in metabolic rates among sites in Marsh Creek in a variety of ways. We used descriptive statistics to summarize means and variation in GPP, ER, and NEP for each site. To assess whether variation in metabolism was predictable based upon site characteristics, we used simple linear regression models of average levels of environmental variables (turbidity, aquatic vegetation, DOC, TP, SRP, TDN, and NO_3^-) to assess their relationships with metabolism estimates (that is, GPP, ER, and NEP). We used only data from May 26, 2017, through October 13, 2017, when we had comparable data for all six reaches. Finally, we used Spearman's rank-order correlation analysis to assess synchrony in metabolism time series between site pairs. We restricted the correlation analysis to the May–October 2017 growing season to enable comparisons of synchrony (quantified as correlation coefficients) among pairs.

Question 2: What Predicts Temporal Variation in Metabolism Within Sites and do those Controls vary Among Sites?

We used two sets of models to evaluate the controls on metabolism and their variation along the study segment. First, we used mixed-effects models (Zuur and others 2009; Harrison and others 2018) to identify the best predictors of metabolic rates along

the whole segment for the period during which we had data from all sites (May–October 2017), including site as a random factor. We then identified the best models for each individual site, using the entire dataset for each site. These latter models enabled us to use all the data available for each site and identify factors that might not have had sufficiently strong effects to be included in the full segment models. In all models, we log-transformed GPP, ER, discharge, and turbidity to meet the assumptions of normality and allow for multiplicative models, which may better represent relationships between metabolism and hypothesized controls.

We built segment-scale mixed-effects models for GPP, ER, and NEP, using the *lme* function of the “nlme” package in R (Pinheiro and others 2020). We used AIC to distinguish between candidate models and an AR1 correlation structure to account for temporal autocorrelation in time series. For GPP, we tested the effects of average daily light (as PAR), temperature, turbidity, discharge, and monthly concentrations of nutrients (NO_3^- , NH_4^+ , TDN, SRP), vegetation cover, and chlorophyll *a* concentration. For ER and NEP, we tested the same predictors as for GPP but also included GPP and dissolved and particulate organic carbon concentrations. We included site as a random factor and compared models in which we allowed only intercepts to vary and models in which we allowed slopes to vary as well. Because overlapping turnover lengths suggest that some site pairs were not independent, we constructed three sets of segment-scale models. We constructed segment-scale models for two subsets of data that turnover lengths suggested were largely independent: Sites 2, 9, and 15, and Sites 2, 8, and 15. We also constructed segment-scale models with all sites included, with the knowledge that these models violate assumptions of independence among sites. Comparisons among these model results enabled us to assess

how robust our full data models were to the violation of the assumption of independence.

For individual site models, we built models for GPP, ER, and NEP for each individual site as well, using the *gl*s function of the “nlme” package in R and using the same predictor variables as above. We used AIC to select among candidate models and an AR1 correlation structure to account for temporal autocorrelation.

RESULTS

Ecosystem structure–water chemistry, aquatic vegetation cover, sestonic chlorophyll *a*, and turbidity–varied substantially along the 75-km segment of Marsh Creek. Some aspects of ecosystem structure, namely sestonic chlorophyll *a* and vegetation cover, were significantly associated with mean turbidity at a site (Figure 2). Rates of ecosystem metabolism varied as much as ecosystem structure (3–4 times among sites) but did not vary systematically with any site characteristics. Controls on metabolism also varied along the segment, and variation in controls was predicted by site-averaged turbidity and total suspended solids.

Variation in Ecosystem Structure Along Marsh Creek

Growing season water chemistry, turbidity, and vegetation cover varied several-fold among sites along Marsh Creek. Some of these parameters were significantly associated with each other. For example, site-averaged turbidity was negatively associated with site-averaged aquatic vegetation cover at a site (Adj. $R^2 = 0.84$, $p = 0.02$) and positively associated with site-averaged sestonic chlorophyll *a* concentration (Adj. $R^2 = 0.86$, $p = 0.01$, Figure 2). Site slope, width, and stream power were unrelated to mean turbidity ($\rho < 0.4$), though there was weak evidence for a positive relationship between slope and total suspended solid concentrations (Adj. $R^2 = 0.56$, $p = 0.09$).

Discharge increased from Site 15 (upstream) to Site 2 (downstream), and seasonal patterns of discharge were similar among sites. A rain-on-snow event in February 2017 resulted in a 5-year flood in Marsh Creek. This flood was followed by the more modest snowmelt-driven increase in discharge in April and seasonal flow recession (Figure S6A). Likewise, water temperature was similar among all sites and followed a predictable seasonal pattern with a winter minimum and summer maximum (Figure S6B). Light generally increased through the summer, though patterns were more variable from

day to day and among sites (Figure S6C). Finally, turbidity was highly variable within and among sites, not appearing to follow any seasonal pattern and with unique temporal patterns for each site (Figure S6D).

Q1. Metabolism was Variable Among Sites Within Marsh Creek, But Unrelated to Site Characteristics

During the growing season (May–October 2017), mean GPP ranged from 1.4 to 4.5 g O₂ m⁻² day⁻¹ (greater than a threefold variation) among sites in Marsh Creek, while ER ranged from – 5.5 to – 1.4 g O₂ m⁻² day⁻¹ (a nearly fourfold variation) among sites (Table 1; Figure 3). Typical for open canopy streams in this region, rates of GPP were generally low during the winter, for sites where we had data, and increased during the summer as temperatures rose and light availability increased (Figure 2). The highest rates of ER occurred during the summer, but ER was also high during the winter. The stream was usually heterotrophic, even during the summer. During the growing season, only 269 out of 755 days across the segment (~ 35%) were autotrophic (NEP > 0; Figure 3). However, growing season trophic status also varied among sites, ranging from 1.4% autotrophic (Site 10) to 79% autotrophic (Site 6, Table 1).

Despite overall similar seasonal patterns of metabolism among sites, there was some variation in the magnitude and timing of the seasonal pattern among sites. Correlation analysis among pairs of sites suggested that GPP was largely synchronous among sites, but that synchrony decayed with distance between sites (Figure 4). Spearman correlation coefficients for GPP ranged from 0.46 to 0.88. ER was less synchronous than GPP overall (Spearman correlation coefficients ranged from – 0.16 to 0.89). Synchrony among sites declined significantly with distance between sites for both ER and GPP (Figure 4), and those declines were greater for ER (ρ decreased by 0.014/km) than for GPP (ρ decreased by 0.006/km), and beyond ~ 20 km, correlation coefficients for ER dropped below 0.4 (Figure 4). NEP was asynchronous among site pairs (Spearman correlation coefficients ranged from – 0.26 to 0.62) and there was no relationship between synchrony and distance between sites (Figure 4). The high correlation coefficients for GPP and ER among Sites 6, 8, 9, and 10 were consistent with overlapping turnover lengths estimated for these sites; yet NEP was only strongly correlated at Sites 8 and 9. The lower correlation for ER and NEP may be related to lower accuracy in estimates of ER

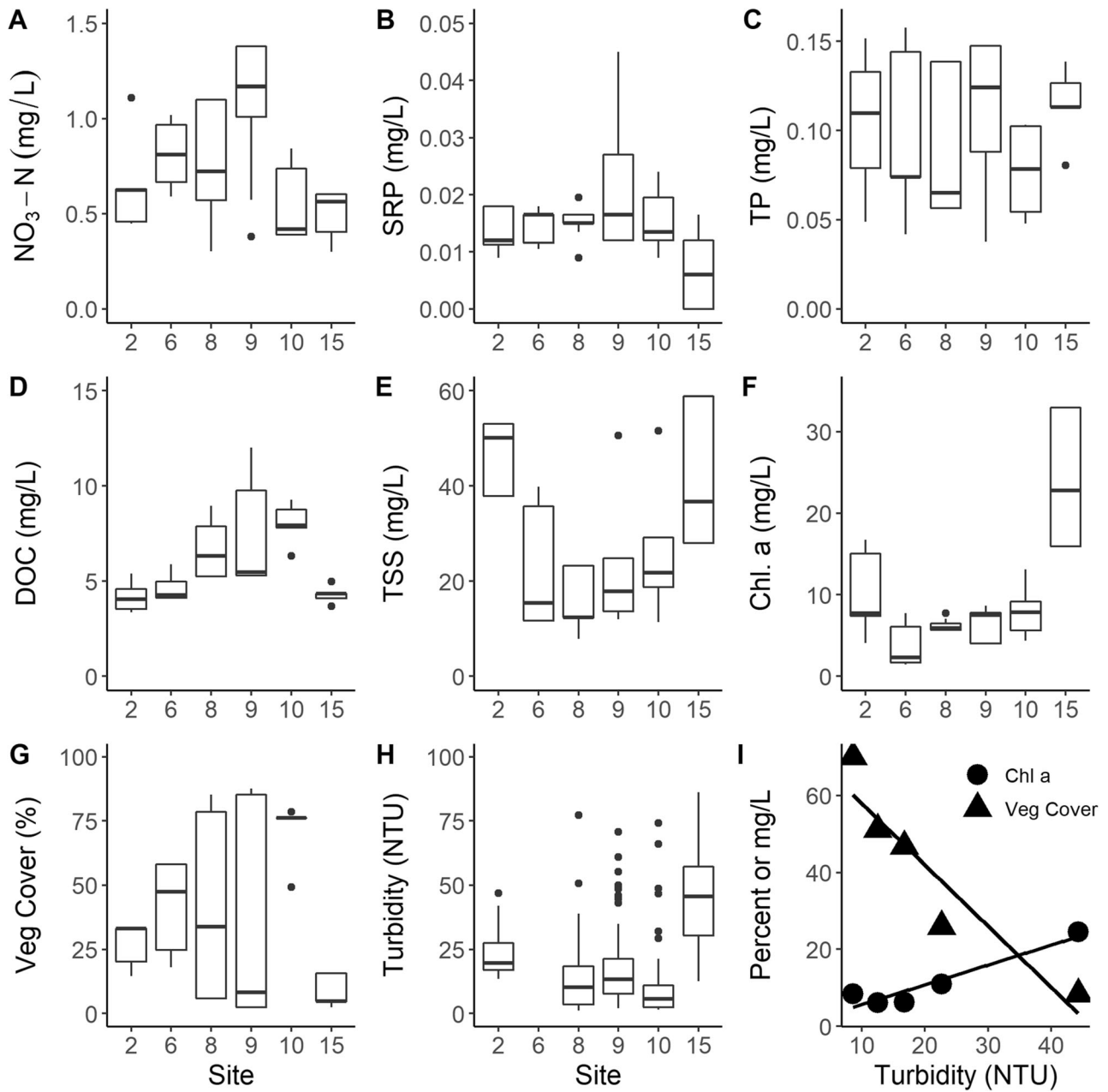


Figure 2. Boxplots of water chemistry (A–E), chlorophyll *a* concentration (F), vegetation cover (G), and turbidity (H) for each site from May to October 2017. I Site-averaged turbidity over the May–October 2017 study period was significantly associated with increases in season-averaged chlorophyll *a* ($R^2 = 0.86$, $p = 0.01$) and decreases in vegetation cover among sites ($R^2 = 0.84$, $p = 0.02$).

compared to GPP. However, if our estimates of ER are accurate, lower correlation of ER among sites could suggest that the broader synchrony in GPP among sites may result from other processes besides non-independence among sites and reflect similar controls on GPP along the stream segment.

We found no relationships between metabolism rates, synchrony among sites, and site characteristics. Growing season average metabolism rates

were uncorrelated to any site characteristics measured in this study, including turbidity, vegetation cover, discharge, water chemistry (nutrients and DOC), and average reach slope (Figure S7). Likewise, synchrony among sites did not appear to be related to similarities in site characteristics or overall metabolic rates. For example, Sites 9 and 15 had the highest rates of GPP but were not synchronous and had quite different site conditions.

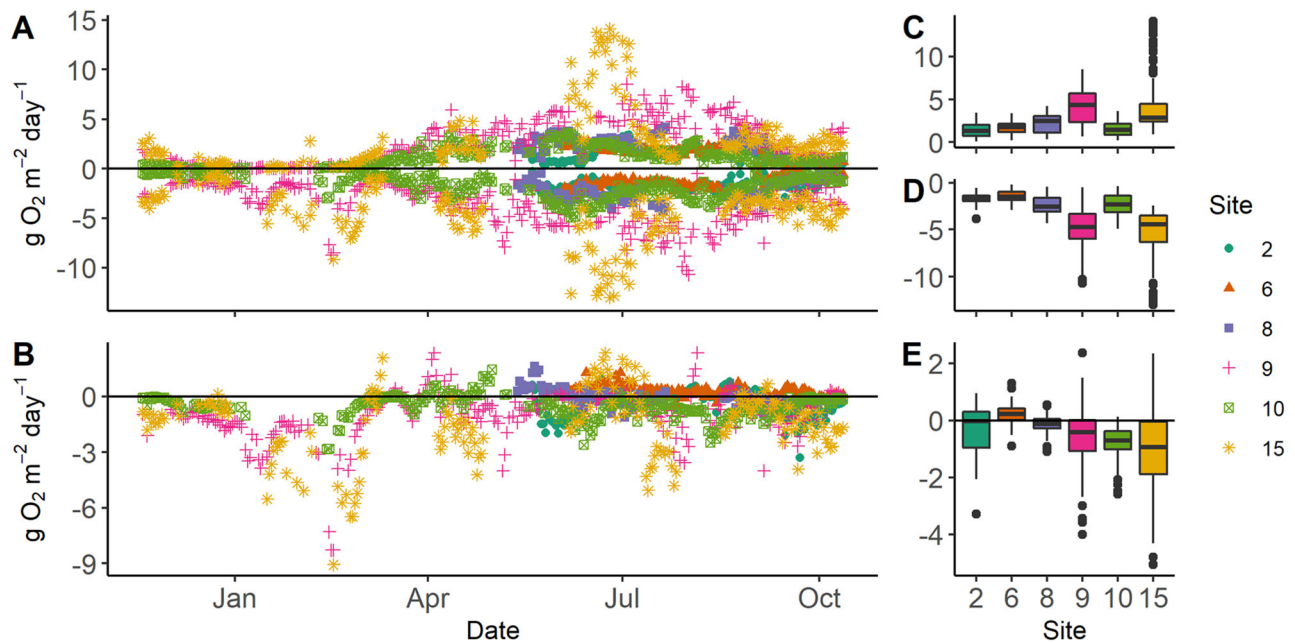


Figure 3. Time series of metabolism. **A** GPP (positive) and ER (negative) and **B** Net ecosystem production (GPP + ER) over full record. Boxplots of **C** GPP, **D** ER, and **E** NEP for each site over the May–October 2017 growing season.

Sites 2, 6, and 10 had lower rates of GPP, but spanned the full range of nutrient concentrations and vegetation cover. Meanwhile, Sites 10 and 15 were more synchronous than other site pairs but had dissimilar water chemistry and rates of metabolism. Overall, these results suggest that rates of GPP and ER are high or low for different reasons along this stream segment.

Q2: Controls on metabolic processes were structured by site characteristics

In segment-scale models, controls on GPP were consistent among sites, whereas ER and NEP had more variable controls along the segment. There were small but important differences in controls on metabolism depending on model scale (segment compared to individual sites). In the best segment-scale models, turbidity was negatively associated with NEP, positively associated with ER, and not associated with GPP. However, individual site models identified turbidity as an important control on GPP, ER, and NEP only for some sites.

Although rates of GPP varied among sites along Marsh Creek, the controls on GPP (stream surface light and water temperature) and their effects were relatively consistent among sites. The best segment-scale model of GPP included only light and temperature, with site as a random factor, and segment-scale model results were robust to variation in included sites, with only small variation in

coefficients (Table 3). Allowing model intercepts to vary for sites improved model performance, reflecting variable rates of GPP among sites. Allowing model slopes to vary among sites also improved model performance, suggesting that GPP at some sites was more responsive to changes in light (Figure S8). However, the differences in GPP–PAR slopes among sites were small (Table 3; Figure S8), and slopes were not significantly related to any measured site characteristics. Although aquatic vegetation cover and soluble reactive phosphorus concentrations did have significant positive effects on GPP in candidate segment-scale models, they were not retained in the best model. This may be in part due to the limited temporal resolution (monthly) of the water chemistry data.

Models of GPP for individual sites were similarly consistent (Table 3). Light and temperature were in a best model for nearly all sites. The best model for Sites 6 and 8 included only light. A major difference between segment- and individual-scale models was the significant negative effect of turbidity on GPP at Sites 2 and 15 (Table 3). Although turbidity did not appear to have a dampening effect on average GPP among sites (Figure S6), it did have a negative effect at the two sites with higher overall turbidity.

GPP was consistently the best predictor of ER among all sites, in both segment-scale and individual site models (Table 4). Allowing the slope of this relationship (in addition to the intercept) to

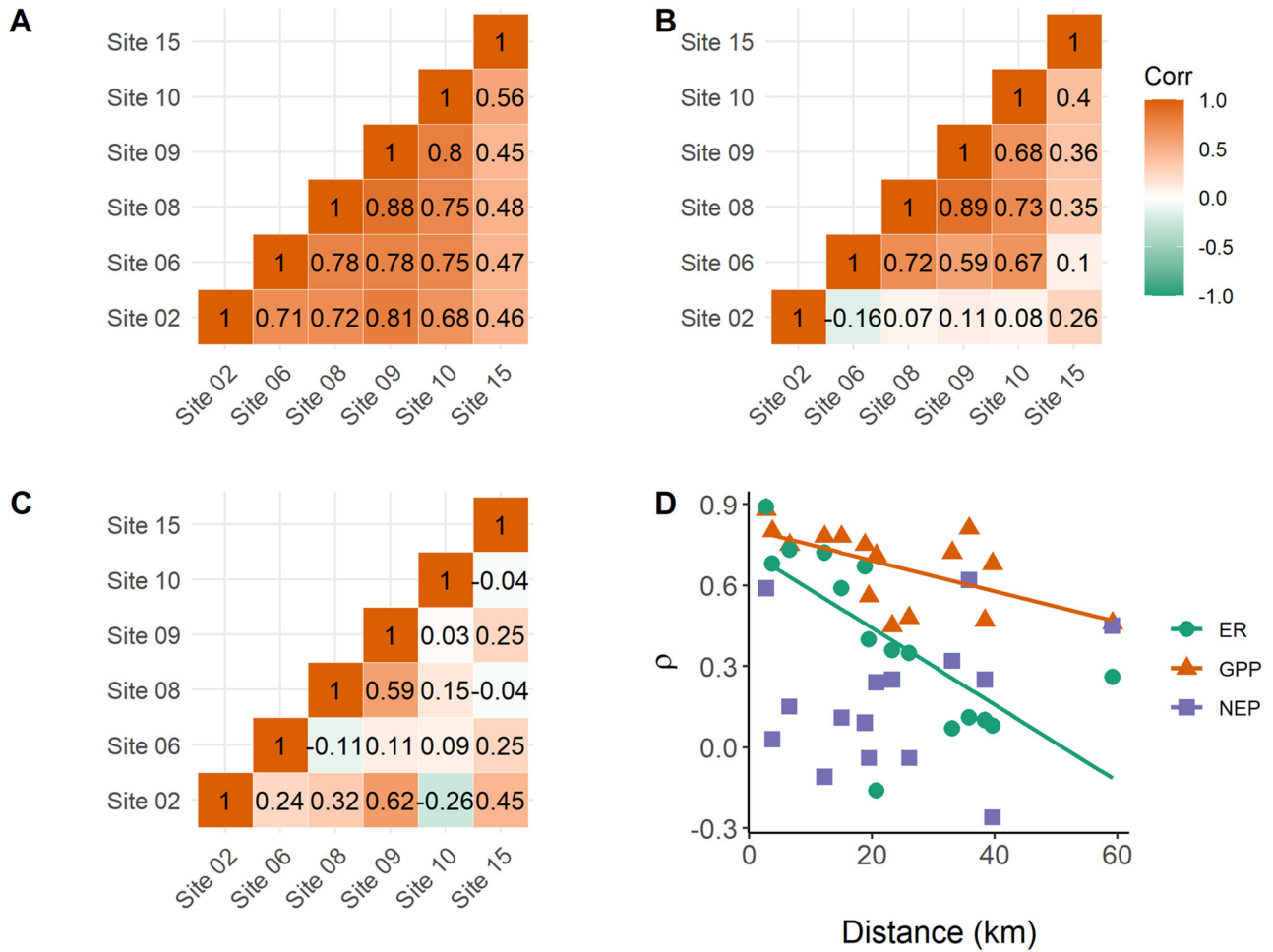


Figure 4. Correlation of metabolism time series among sites. **A** GPP, **B** ER, **C** NEP. **D** Decay in metabolic synchrony (Spearman's rho) with distance between sites (GPP: Adj. $R^2 = 0.30$, $p = 0.02$, ER: Adj. $R^2 = 0.46$, $p = 0.003$, NEP: Adj. $R^2 = 0$, $p = 0.52$).

Table 3. Parameter Estimates for the Best-Fit Models for Log(GPP) for the Whole Segment (Sites 2, 9, and 15; Sites 2, 8, and 15; and All Sites) with Site as a Random Factor and Individual Sites

Site	Intercept	Log (Turbidity)	Light	Temp
Segment (Sites 2, 9, and 15)	0.1	—	0.0003**	0.022**
Segment (Sites 2, 8, and 15)	0.09	—	0.0004**	0.015**
Segment (all sites)	0.09	—	0.0003**	0.018**
2	0.43	— 0.32**	0.0002**	0.015**
6	0.26	—	0.0005**	—
8	0.32	—	0.0004**	—
9	0.09	—	0.0003**	0.030**
10	0.07	—	0.0003**	0.015**
15	0.43	— 0.21**	0.0005**	0.025**

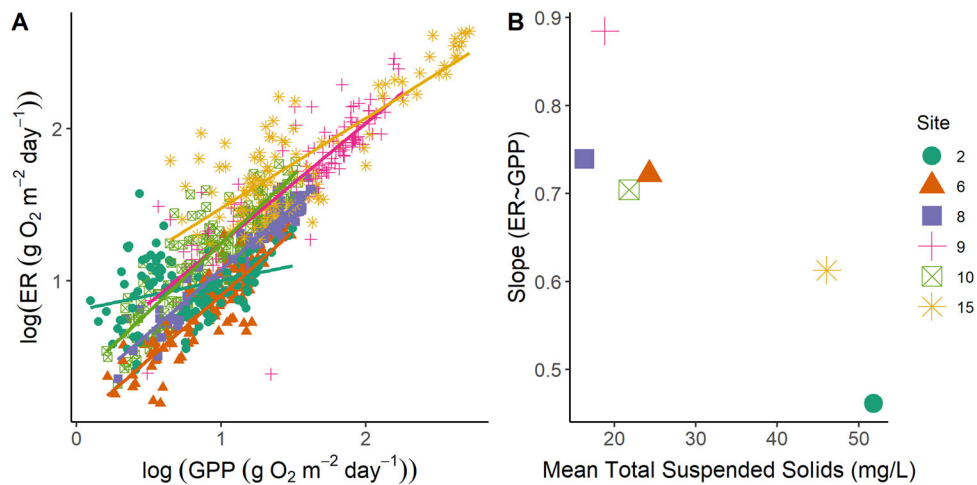
* $0.05 < p < 0.1$; ** $p < 0.05$.

vary among sites substantially improved model performance ($\Delta AIC > 2$). Although GPP was the best predictor among sites, the strength of that

relationship varied significantly among sites. Furthermore, the slope of the GPP–ER relationship was strongly related to site-averaged total suspended

Table 4. Parameter Estimates for the Best-Fit Models for $\text{Log}(-\text{ER})$ for the Whole Segment (Sites 2, 9, and 15; Sites 2, 8, and 15; and All Sites) with Site as a Random Factor, and Individual Sites

Site	Intercept	GPP	Log (Turbidity)	Log (Discharge)
Segment (Sites 2, 9, and 15)	0.06	0.74**	0.12**	–
Segment (Sites 2, 8, and 15)	0.18	0.60**	–	0.30**
Segment (all sites)	0.2	0.71**	–	–
2	– 0.51	0.26**	0.29**	0.58**
6	0.06	0.33**	–	–
8	0.23	0.34**	–	0.35**
9	0.25	0.34**	–	–
10	0.18	0.31**	–	–
15	0.15	0.2**	0.13**	0.49**
15	0.35	0.2**	–	0.52**

* $0.05 < p < 0.1$; ** $p < 0.05$.**Figure 5.** **A** Variation in ER–GPP relationships among sites (ER is expressed in positive values to enable log transformation for analysis), and **B** GPP–ER slope against mean TSS. Adj. $R^2 = 0.77$, $p = 0.01$. Data are from May to October 2017 only. Note that slopes shown in **(B)** account for temporal autocorrelation, whereas those in panel **(A)** do not and are shown for illustration.

solid (TSS) concentrations, such that ER was more strongly related to GPP at sites with lower TSS concentrations (Adj. $R^2 = 0.77$, $p = 0.02$, Figure 5).

Turbidity had a significant positive effect on ER in all segment-scale models but was retained in only one of the best segment-scale models (Table 4). In the individual site models, turbidity had a significant positive effect on ER and was retained in the best models for Sites 2 and 15 (Table 4).

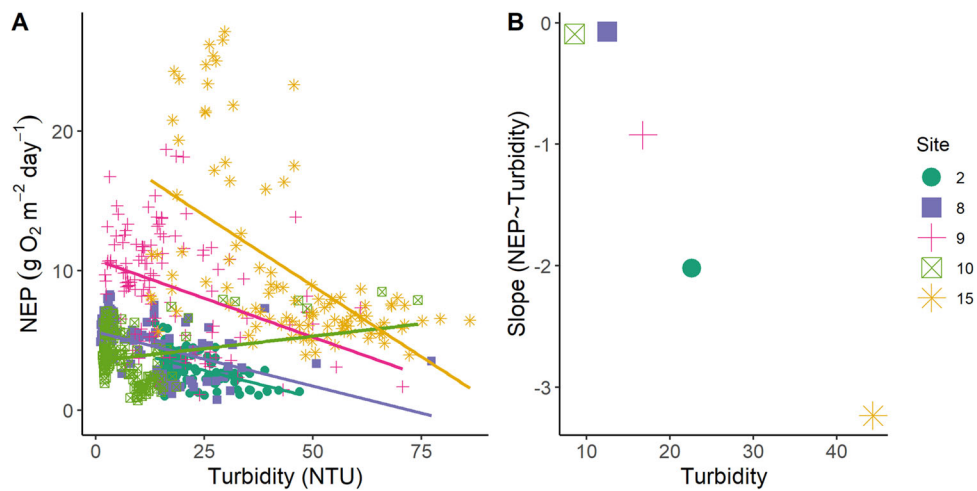
Unlike GPP and ER, turbidity had a significant negative effect on NEP in all segment-scale models (Table 5). The direction of the effect was similar among segment-scale models, but the magnitude of the effect varied by a factor of 2.3, reflecting variation in the effect of turbidity on NEP along the segment. NEP was also positively associated with light at the segment scale, and light was a consis-

tent predictor of NEP in the individual site models (Table 5). Reflecting the contribution of GPP to NEP, the light–NEP relationship was consistent among sites in the segment-scale model and individual site models (Table 5). Discharge (positive effect) and turbidity (negative effect) were included in some individual site models (Table 5).

Allowing the slope of the turbidity–NEP relationship (in addition to the intercept) to vary among sites in the segment-scale model improved model performance. Reflecting otherwise consistent controls on NEP along the segment, turbidity was the only variable for which allowing slopes to vary improved the model. Furthermore, the slope of the NEP–turbidity relationship was significantly related to site-averaged turbidity (Adj. $R^2 = 0.90$, $p = 0.01$), such that turbidity had a stronger nega-

Table 5. Parameter Estimates for the Best-Fit Models for NEP for the Whole Segment (Sites 2, 9, and 15; Sites 2, 8, and 15; and All Sites), With Site as a Random Factor, and Individual Sites

Site	Intercept	Log (Turbidity)	Light	Log (Discharge)
Segment (Sites 2, 9, and 15)	0.3	− 1.3**	0.0026**	—
Segment (Sites 2, 8, and 15)	− 0.41	− 0.83**	0.0025**	—
Segment (all sites)	0.66**	− 0.56**	0.0022**	—
2	3.1	− 1.39**	0.002**	− 4.5**
6	− 0.22	—	0.001**	—
8	− 1.4**	—	0.001**	2.4**
9	− 0.86	—	0.002**	− 1.3*
10	− 0.81**	—	0.001**	—
15	1.8	− 2.16**	0.003**	− 4.6**

* $0.05 < p < 0.1$; ** $p < 0.05$.**Figure 6.** **A** Variation in turbidity–NEP relationships among sites. **B** The slope of the NEP–turbidity relationship from the segment-scale mixed-effects model is significantly correlated with site-averaged turbidity ($R^2 = 0.90$, $p = 0.01$). Data are from May to October 2017. Note that slopes shown in **(B)** account for temporal autocorrelation and the effects of light, whereas those in panel **(A)** do not and are shown for illustration.

tive effect at sites with higher average turbidity, and little to no effect at sites with low average turbidity (Figure 6). As with GPP and ER, we found that turbidity was retained in the best individual site models only for the most turbid sites, Sites 2 and 15.

DISCUSSION

We found evidence of distinct process domains (Montgomery 1999) along Marsh Creek that are likely delineated by complex interactions between geomorphological and ecological processes. Instead of a simple relationship between geomorphology and metabolism, turbidity and suspended solids were likely affected by a range of geomorphic processes (for example, erosion, deposition, resus-

pension) that themselves varied along the stream segment, and which in turn affected the controls on ecosystem metabolism. At turbid sites with high suspended solids, GPP and NEP were depressed by turbidity, while ER was relatively decoupled from GPP and positively associated with turbidity. At less turbid sites, GPP was unaffected by turbidity and ER and GPP were tightly coupled. Importantly, while turbidity structured the *controls* on metabolism, it did not appear to affect the overall *rates* of metabolism. We hypothesize that cycles of sediment deposition and resuspension alters the distribution of particle sizes, bacterial communities associated with particulate matter, the quality of dissolved organic matter, and concentrations of particulate nutrients and organic matter in ways that would influence ecosystem process rates,

obscuring any direct relationships between turbidity and metabolism rates among sites.

Spatial Variation in Metabolism Unrelated to Site Characteristics

Site-averaged GPP and ER varied greater than threefold along Marsh Creek. The variation observed here was higher than that observed along agricultural streams (Griffiths and others 2013), but relatively low compared to the greater than tenfold variation observed in natural stream segments (Dodds and others 2018) and networks (Dodds and others 2018; Mejia and others 2018; Ulseth and others 2018). Daily variation within Marsh Creek was similar to spatial variation in site-averaged metabolism reported by Appling and others (2018b), though average rates within Marsh Creek sites (GPP: $2.1 \text{ g O}_2 \text{ m}^{-2} \text{ day}^{-1}$, ER: $-2.6 \text{ g O}_2 \text{ m}^{-2} \text{ day}^{-1}$) were slightly below the mean from the Appling and others (2018b) dataset (GPP: $2.7 \text{ g O}_2 \text{ m}^{-2} \text{ day}^{-1}$; ER: $-4.4 \text{ g O}_2 \text{ m}^{-2} \text{ day}^{-1}$) and below the mean metabolism observed in agricultural streams (Bernot and others 2010, GPP: $4 \text{ g O}_2 \text{ m}^{-2} \text{ day}^{-1}$, ER: $-6 \text{ g O}_2 \text{ m}^{-2} \text{ day}^{-1}$) (Supplemental Figure S22). Rates were also much lower than mean spring rates observed in another agricultural stream, Shatto Ditch, in Indiana (Roley and others 2014, GPP: $4.5 \text{ g O}_2 \text{ m}^{-2} \text{ day}^{-1}$, ER: $-6.1 \text{ g O}_2 \text{ m}^{-2} \text{ day}^{-1}$).

When analyzed across sites at the scale of the entire segment, seasonally averaged metabolism rates in Marsh Creek were unrelated to site conditions, despite a similar degree of variation in water chemistry, primary producers, and turbidity. Spatial variation in metabolic process rates (usually measured over relatively short time periods) is usually correlated with site characteristics such as light, nutrient concentrations, temperature, turbidity (Izagirre and others 2008; Bernot and others 2010; Mejia and others 2018; Kaylor and others 2019), though this is not always the case (Fuß and others 2017; Dodds and others 2018). In Marsh Creek, the lack of a relationship between site conditions and metabolism could be due to three reasons: (1) site conditions that were measured either at the sensor (water chemistry) or for 100-m upstream of the sensor (vegetation cover) may not spatially represent site conditions for metabolism that is integrated for hundreds of meters upstream of the sensor; (2) variation in the quality of particulate matter and turbidity among sites (for example, particle size distribution, particulate nutrients, quality of organic matter, microbial loads associated with particulates) may have obscured

simple correlations between turbidity and metabolism (for example, see Figure S9); and/or (3) relationships may be obscured by noise associated with uncertainty in discharge and depth, and therefore metabolism estimates.

Water chemistry is frequently measured at the sensor location to understand linkages between water chemistry and metabolism. Yet if nutrients are expected to control metabolism, measuring concentrations at the sensor locations provides information about nutrient concentrations *after* they have been affected by metabolism and uptake, rather than information about the nutrients that were available for uptake across the reach. Understanding site-level controls on metabolism may require sampling water chemistry at the upstream end of a study reach (ideally using a two-station method to estimate metabolism that would enable confident delineation of the reach extent). Otherwise, sampling at the downstream end of the reach confounds the effects of nutrients on metabolism (Peterson and others 1985; Bernot and others 2010; Yates and others 2013; Dodds and others 2018; Kominoski and others 2018) and the effects of metabolism on nutrients (Roberts and others 2007; Heffernan and Cohen 2010; Cohen and others 2013).

Turbidity can lower GPP by blocking light (Griffiths and others 2013; Hall and others 2015; Dodds and others 2018). Yet, suspended solids are often associated with nutrients, which could have a stimulatory effect on GPP (Summers and others 2020), and organic matter and microbial loads could stimulate respiration (Ulseth and others 2018; Summers and others 2020). Therefore, differences in the quality of particulate matter among sites could obscure correlations between turbidity and metabolism. Better characterization of particulate materials, not just organic and nutrient concentrations, but also the quality of organic matter, particle size distributions, and microbial loads, may better explain spatial variation in metabolism along Marsh Creek. For example, the proportion of total suspended solids that were organic (AFDM/TSS) varied from below detection limit to 0.6 among sites and months, with Site 2 having approximately half the amount of organic matter on average compared to other sites (Figure S9). Unfortunately, fully disentangling the numerous and interacting controls on stream metabolism requires both large numbers of sites and long datasets at each site to provide the needed statistical power to evaluate these relationships.

Temporal Patterns and Overall Drivers of Metabolism in Marsh Creek

At all sites along Marsh Creek, GPP and ER generally peaked in mid- to late summer, characteristic of open canopy streams (Savoy and others 2019). Where we had data, we observed some of the highest rates of ER during early spring (February/March), likely due to major flooding that occurred during this time (unfortunately, the flooding was one reason we did not have data for all sites during this period). Indeed, ER at several sites and in one segment-scale model was positively associated with discharge (Table 4). In general, DOC is flushed into streams with high flows (Raymond and others 2016), and high DOC fluxes may also be accompanied by high bacterial loads from soils (Caillon and Schelker 2020). Together, these can fuel heterotrophic respiration (Ulseth and others 2018; Demars 2019; Summers and others 2020).

Along Marsh Creek as a whole, light and temperature emerged as the only significant predictors of GPP. Surprisingly, despite highly variable turbidity within and among sites, turbidity did not emerge as a control on GPP in segment-scale models. This result contrasts with studies that have found a dampening effect of turbidity on GPP (Izagirre and others 2008; Griffiths and others 2013), even in low turbidity conditions (Hosen and others 2014). We also did not find any relationships between nutrients or vegetation cover and GPP, although there were significant effects of SRP and vegetation cover that were not retained in the best models. In previous studies of Marsh Creek, biofilm primary producers were limited by nitrogen and phosphorous during parts of the year (Marcarelli and others 2009; Bechtold and others 2012), so our results are likely because of the coarse resolution of our nutrient data (monthly), rather than a lack of nutrient limitation in Marsh Creek. Higher frequency nutrient sampling is likely required to identify relationships between metabolism and nutrients within sites (Heffernan and Cohen 2010; Cohen and others 2013).

Overall, ER was strongly predicted by GPP in Marsh Creek, given the large proportion of respiration by autotrophs common in stream ecosystems (Griffiths and others 2013; Mejia and others 2018; Ulseth and others 2018; but also see Kaylor and others 2019). Unlike other studies (Lamberti and Steinman 1997; Mulholland and others 2001; Izagirre and others 2008; Bernot and others 2010; Kaylor and others 2019), we found no effects of turbidity, temperature, or nutrients when we analyzed ER across sites. NEP is only rarely mod-

eled as an independent response variable (Dodds and others 2018; Ulseth and others 2018). In Marsh Creek, NEP increased with light and decreased with turbidity. The negative effect of turbidity emerged for NEP likely because of a positive effect of turbidity on ER (Izagirre and others 2008; Dodds and others 2018) and negative effect on GPP (Griffiths and others 2013; Hall and others 2015) that combined to create a more substantial negative effect on NEP that was large enough to be identified in the model.

Variable Drivers of Metabolism Along Marsh Creek Reveal Process Domains

Rates of GPP, ER, and NEP had different controls among sites that were significantly related to site-averaged turbidity and concentrations of total suspended solids (TSS). Most studies on the controls on metabolism compare rates among sites, relating these to site characteristics (Bernot and others 2010; Dodds and others 2018; Kaylor and others 2019). Fewer, but increasing, studies address drivers of temporal variation within a site (Roberts and others 2007; Griffiths and others 2013; Summers and others 2020), and even fewer compare temporal patterns among sites (Mejia and others 2018; Ulseth and others 2018; Savoy and others 2019; Myrstener and others 2021). Our work shows that the controls on metabolism can vary along a stream segment in ways that are predicted by site characteristics.

Streams are geomorphically patchy ecosystems (Montgomery 1999; Poole 2002), and that structural patchiness leads to variation in ecosystem process rates (Bellmore and Baxter 2014; Dodds and others 2018) and the controls on those rates (Thorp and others 2006). Despite a lack of major geomorphological transitions, we found evidence that process domains exist within this segment of Marsh Creek, suggesting that the linkages between stream ecology and geomorphology may be more complex. Marsh Creek has an exaggerated step/pool morphology, resulting from both human manipulation (numerous low-head dams constructed for irrigation) and geomorphic legacy (relict cobble boulder bars from the Bonneville Flood (O'Connor 1993)). This step/pool morphology regulates sediment deposition and entrainment, thus observed longitudinal variability in turbidity (Guilinger 2017; Meese 2018). These irregular patterns of turbidity, in turn, apparently determined the controls on metabolism within each reach.

Reaches along this segment of Marsh Creek represent different process domains with respect to metabolic processes: More turbid reaches see negative effects of turbidity on GPP and positive effects on ER. Because turbidity is also associated with increases in particulate and dissolved organic carbon, total phosphorus, and microbial loads (Young and Huryn 1996; Izaguirre and others 2008; Caillon and Schelker 2020), turbidity is associated with increased respiration and decoupled ER and GPP (Biggs and others 1999; Uehlinger and others 2002; Atkinson and others 2008). Other, less turbid sites have stronger coupling between GPP and ER, and GPP is predicted by light and temperature alone.

Our ecological analysis suggests the presence of process domains: variation in the controls on ecological processes, and the role of turbidity and suspended solids suggest that these are delineated by geomorphic processes. Typically, process domains (or functional process zones) are first delineated and then used to explain ecological patterns, with varying success (for example, Bellmore and Baxter 2014; Collins and others 2018). Yet a fundamental principle in landscape ecology is that the relevant landscape structure and patterns depend on the ecological phenomenon of interest. Thus, we suggest that a more fruitful pathway for identifying process domains may be to work from the ecological process to identify what geomorphic processes and features are important. These are likely to vary depending on the scale and process of interest.

Unfortunately, we cannot with this dataset identify the boundaries or the geomorphic definitions of the process domains observed in this study. Our inability to do so is largely because how metabolism is typically measured, with a one-station approach, is ill-equipped for the task of linking ecosystem process to geomorphic structure. The one-station method yields estimates of process rates along a long and variable reach extent that integrates over substantial ecological (Dodds and others 2018) and geomorphic heterogeneity (for example, Figure 1C). The averaging effect of the single-station metabolism approach may be useful for capturing broad patterns of metabolism, but limits our ability to understand the controls on metabolism and link these to geomorphic variation. Delineating the spatial boundaries of process domains would require a two-station approach to bound metabolism estimates to a specific reach length, which could be targeted to capture hypothesized variation in geomorphic structure and function. That said, the two-station approach is not without its chal-

lenges, including a relatively large minimum scale and the need for accurate travel time estimates.

Just as short-term measures can misrepresent annual metabolism (Roberts and others 2007; Griffiths and others 2013), measurement of a single reach within a stream segment may misrepresent both the magnitude of metabolic activity as well as the controls on that activity (Van de Bogert and others (2012) show similar challenges in lakes). This finding is an especially important consideration for efforts to use single-site measurements to predict the future metabolic state of rivers (for an earlier attempt from our study region, see Marcarelli and others 2010) or understand patterns of metabolism across biomes (Bernhardt and others 2017; Savoy and others 2019). Moreover, what may be important to the metabolism of a river as a whole may not be the metabolic character of any one process domain, but the relationships among these process domains, longitudinally and in networks (Bellmore and Baxter 2014; Mejia and others 2018; Kaylor and others 2019).

CONCLUSION

Metabolism is highly variable over space and time (Bernhardt and others 2017; Dodds and others 2018; Savoy and others 2019), and the controls on that variability can themselves vary depending on study extent and resolution as well as seasonally within a reach. Stream metabolism does not respond to environmental controls (for example, light) the same way among ecosystems, in part because these controls do not operate independently. Rather, metabolism is determined by the interactions among controls, and some of these interactions act across scales. Unfortunately, the complex models needed to fully evaluate the numerous and interacting controls on stream metabolism require both large numbers of sites and long datasets at each site. Our work illustrates the potential for long term data at many sites to elucidate multi-level controls on metabolism. Quantifying how geomorphological variation across scales structures the controls on ecosystem function within stream networks will be fundamental for making predictions about metabolism, and therefore carbon cycling, among streams.

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DATA AVAILABILITY

Honious, S.A., R.L. Hale, J.J. Guilinger, B.T. Crosby, and C.V. Baxter. 2021. Stream Metabolism in Marsh Creek, Idaho, USA 2016–2017 ver 1. Environmental Data Initiative. <https://doi.org/10.6073/pasta/c5dcdde09dbbee91764e3e6f5ee81696>

REFERENCES

- Appling AP, Hall RO, Yackulic CB, Arroita M. 2018a. Overcoming equifinality: leveraging long time series for stream metabolism estimation. *Journal of Geophysical Research: Biogeosciences* 123:624–645.
- Appling AP, Read JS, Winslow LA, Arroita M, Bernhardt ES, Griffiths NA, Hall RO, Harvey JW, Heffernan JB, Stanley EH, Stets EG, Yackulic CB. 2018b. The metabolic regimes of 356 rivers in the United States. *Scientific Data* 5:180292.
- Atkinson BL, Grace MR, Hart BT, Vanderkruk KEN. 2008. Sediment instability affects the rate and location of primary production and respiration in a sand-bed stream. *Journal of the North American Benthological Society* 27:581–592.
- Battin TJ, Kaplan LA, Findlay S, Hopkinson CS, Marti E, Packman AI, Newbold JD, Sabater F. 2008. Biophysical controls on organic carbon fluxes in fluvial networks. *Nature Geoscience* 1:95–100.
- Battin TJ, Luyssaert S, Kaplan LA, Aufdenkampe AK, Richter A, Tranvik LJ. 2009. The boundless carbon cycle. *Nature Geoscience* 2:598–600.
- Bechtold HA, Marcarelli AM, Baxter CV, Inouye RS. 2012. Effects of N, P, and organic carbon on stream biofilm nutrient limitation and uptake in a semi-arid watershed. *Limnology and Oceanography* 57:1544–1554.
- Bellmore JR, Baxter CV. 2014. Effects of geomorphic process domains on river ecosystems: a comparison of floodplain and confined valley segments. *River Research and Applications* 30:617–630.
- Benda L, Poff NL, Miller D, Dunne T, Reeves G, Pess G, Pollock M. 2004. The Network Dynamics Hypothesis: How Channel Networks Structure Riverine Habitats. *BioScience* 54:413.
- Bernhardt ES, Heffernan JB, Grimm NB, Stanley EH, Harvey JW, Arroita M, Appling AP, Cohen MJ, McDowell WH, Hall RO, Read JS, Roberts BJ, Stets EG, Yackulic CB. 2017. The metabolic regimes of flowing waters. *Limnology and Oceanography* 63:599–118.
- Bernot MJ, Sobota DJ, Hall RO, Mulholland PJ, Dodds WK, Webster JR, Tank JL, Ashkenas LR, Cooper LW, Dahm CN, Gregory SV, Grimm NB, Hamilton SK, Johnson SL, McDowell WH, Meyer JL, Peterson B, Poole GC, Maurice Valett HM, Arango C, Beaulieu JJ, Burgin AJ, Crenshaw C, Helton AM, Johnson L, Merriam J, Niederlehner BR, O'Brien JM, Potter JD, Sheibley RW, Thomas SM, Wilson K. 2010. Inter-regional comparison of land-use effects on stream metabolism. *Freshwater Biology* 55:1874–1890.
- Biggs BJF, Smith RA, Duncan MJ. 1999. Velocity and sediment disturbance of periphyton in headwater streams: biomass and metabolism. *Journal of the North American Benthological Society* 18:222–241.
- Bowden WB, Glime JM, Riis T. 2007. Macrophytes and Bryophytes. In: *Methods in Stream Ecology*. pp 381–414.
- Caillon F, Schelker J. 2020. Dynamic transfer of soil bacteria and dissolved organic carbon into small streams during hydrological events. *Aquatic Sciences* 82:41.
- Cohen MJ, Kurz MJ, Heffernan JB, Martin JB, Douglass RL, Foster CR, Thomas RG. 2013. Diel phosphorus variation and the stoichiometry of ecosystem metabolism in a large spring-fed river. *Ecological Monographs* 83:155–176.
- Cole JJ, Prairie YT, Caraco NF, McDowell WH, Tranvik LJ, Striegl RG, Duarte CM, Kortelainen P, Downing JA, Middelburg JJ, Melack J. 2007. Plumbing the global carbon cycle: integrating inland waters into the terrestrial carbon budget. *Ecosystems* 10:172–185.
- Collins SE, Matter SF, Buffam I, Flotemersch JE. 2018. A patchy continuum? Stream processes show varied responses to patch- and continuum-based analyses. *Ecosphere* 9:e02481.
- Demars BOL. 2019. Hydrological pulses and burning of dissolved organic carbon by stream respiration. *Limnology and Oceanography* 64:406–421.
- Demars BOL, Thompson J, Manson JR. 2015. Stream metabolism and the open diel oxygen method: principles, practice, and perspectives. *Limnology and Oceanography: Methods* 13:356–374.
- de Vente J, Poesen J, Arabkhedri M, Verstraeten G. 2007. The sediment delivery problem revisited. *Progress in Physical Geography* 31:155–178.
- Dodds WK, Gido K, Whiles MR, Daniels MD, Grudzinski BP. 2015. The Stream Biome Gradient Concept: factors controlling lotic systems across broad biogeographic scales. *Freshwater Science* 34:1–19.
- Dodds WK, Higgs SA, Spangler MJ, Guinnip J, Scott JD, Hedden SC, Frenette BD, Taylor R, Schechner AE, Hoeinghaus DJ, Evans-White MA. 2018. Spatial heterogeneity and controls of ecosystem metabolism in a Great Plains river network. *Hydrobiologia* 813:85–102.
- Drake TW, Raymond PA, Spencer RGM. 2017. Terrestrial carbon inputs to inland waters: a current synthesis of estimates and uncertainty. *Limnology and Oceanography Letters* 13:2–42.
- Fuß T, Behounek B, Ulseth AJ, Singer GA. 2017. Land use controls stream ecosystem metabolism by shifting dissolved organic matter and nutrient regimes. *Freshwater Biology* 62:582–599.
- Griffiths NA, Tank JL, Royer TV, Roley SS, Rosi-Marshall EJ, Whiles MR, Beaulieu JJ, Johnson LT. 2013. Agricultural land use alters the seasonality and magnitude of stream metabolism. *Limnology and Oceanography* 58:1513–1529.
- Guilinger JJ. 2017. Appraisal of suspended sediment sources and fluxes in Marsh Creek, SE Idaho: complexities revealed by high-resolution spatial and temporal data. Master's Thesis. Idaho State University. Pocatello, ID.
- Hall RO. 2016. Chapter 4 - Metabolism of Streams and Rivers: Estimation, Controls, and Application. In: Jones JB, Stanley EH, Eds. *Stream Ecosystems in a Changing Environment*. Boston: Academic Press. pp 151–180.

- Hall RO, Yackulic CB, Kennedy TA, Yard MD, Rosi-Marshall EJ, Voichick N, Behn KE. 2015. Turbidity, light, temperature, and hydropeaking control primary productivity in the Colorado River, Grand Canyon. *Limnology and Oceanography* 60:512–526.
- Harrison XA, Donaldson L, Correa-Cano ME, Evans J, Fisher DN, Goodwin CED, Robinson BS, Hodgson DJ, Inger R. 2018. A brief introduction to mixed effects modelling and multi-model inference in ecology. *PeerJ* 6:e4794.
- Heffernan JB, Cohen MJ. 2010. Direct and indirect coupling of primary production and diel nitrate dynamics in a subtropical spring-fed river. *Limnology and Oceanography* 55:677–688.
- Hosen JD, McDonough OT, Febria CM, Palmer MA. 2014. Dissolved organic matter quality and bioavailability changes across an urbanization gradient in headwater streams. *Environmental Science and Technology* 48:7817–7824.
- Hotchkiss ER, Hall RO Jr, Sponseller RA, Butman D, Klaminder J, Laudon H, Rosvall M, Karlsson J. 2015. Sources of and processes controlling CO₂ emissions change with the size of streams and rivers. *Nature Geoscience* 8:696–699.
- IASCD (Idaho Association of Soil Conservation Districts). 2013. Marsh Creek Monitoring Project. Pocatello, Idaho.
- IDEQ (Idaho Department of Environmental Quality). 2010. Portneuf River Subbasin Assessment and TMDLs Revision and Addendum. Pocatello, Idaho.
- Izagirre O, Agirre U, Bermejo M, Pozo J, Elozegi A. 2008. Environmental controls of whole-stream metabolism identified from continuous monitoring of Basque streams. *Journal of North American Benthological Society* 27:252–268.
- Jankowski K, Schindler DE, Lisi PJ. 2014. Temperature sensitivity of community respiration rates in streams is associated with watershed geomorphic features. *Ecology* 95:2707–2714.
- Jankowski KJ, Schindler DE. 2019. Watershed geomorphology modifies the sensitivity of aquatic ecosystem metabolism to temperature. *Scientific Reports* 9:17619.
- Kaylor MJ, White SM, Saunders WC, Warren DR. 2019. Relating spatial patterns of stream metabolism to distributions of *Juveniles salmonids* at the river network scale. *Ecosphere* 10:e02781.
- Kominoski JS, Rosemond AD, Benstead JP, Gulis V, Manning DWP. 2018. Experimental nitrogen and phosphorus additions increase rates of stream ecosystem respiration and carbon loss. *Limnology and Oceanography* 63:22–36.
- Lamberti GA, Steinman AD. 1997. A comparison of primary production in stream ecosystems. *Journal of the North American Benthological Society* 16:95–104.
- Long MH, Rheuban JE, Berg P, Zieman JC. 2012. A comparison and correction of light intensity loggers to photosynthetically active radiation sensors. *Limnology and Oceanography: Methods* 10:416–424.
- Marcarelli AM, Bechtold HA, Rugenski AT, Inouye RS. 2009. Nutrient limitation of biofilm biomass and metabolism in the Upper Snake River basin, southeast Idaho, USA. *Hydrobiologia* 620:63–76.
- Marcarelli AM, Kirk RWV, Baxter CV. 2010. Predicting effects of hydrologic alteration and climate change on ecosystem metabolism in a western U.S. river. *Ecological Applications* 20:2081–2088.
- McSorley MR. 1977. Water Quality Studies: Marsh Creek, Portneuf River, Bear River and Blackfoot River: Bannock and Caribou Counties. Idaho Department of Health & Welfare, Division of Environment.
- Meese GA. 2018. Farmers, Fences, and Fine Sediment: Using Long-Term Water Quality Data and Aerial Imagery to Quantify the Impact of Best Management Practices in Marsh Creek, ID. Master's Thesis. Idaho State University. Pocatello, ID.
- Mejia FH, Fremier AK, Benjamin JR, Bellmore JR, Grimm AZ, Watson GA, Newsom M. 2018. Stream metabolism increases with drainage area and peaks asynchronously across a stream network. *Aquatic Sciences* 81:9.
- Minshall GW, Petersen RC, Cummins KW, Bott TL, Sedell JR, Cushing CE, Vannote RL. 1983. Interbiome comparison of stream ecosystem dynamics. *Ecological Monographs* 53:2–25.
- Montgomery DR. 1999. Process domains and the river continuum. *Journal of the American Water Resources Association* 35:397–410.
- Mulholland PJ, Fellows CS, Tank JL, Grimm NB, Webster JR, Hamilton SK, Marti E, Ashkenas L, Bowden WB, Dodds WK, McDowell WH, Paul MJ, Peterson BJ. 2001. Inter-biome comparison of factors controlling stream metabolism. *Freshwater Biology* 46:1503–1517.
- Myrstener M, Gómez-Gener L, Rocher-Ros G, Giesler R, Sponseller RA. 2021. Nutrients influence seasonal metabolic patterns and total productivity of Arctic streams. *Limnology and Oceanography* 66:S182–S196.
- O'Connor JE. 1993. Hydrology, Hydraulics, and Geomorphology of the Bonneville Flood. <https://pubs.geoscienceworld.org/books/book/410/chapter/3798074/Hydrology-Hydraulics-and-Geomorphology-of-the>. Last accessed 04/01/2021
- Odum HT. 1956. Primary production in flowing waters. *Limnology and Oceanography* 1:102–117.
- Peterson BJ, Hobbie JE, Hershey AE, Lock MA, Ford TE, Vestal JR, McKinley VL, Hullar M a. J, Miller MC, Ventullo RM, Volk GS. 1985. Transformation of a tundra river from heterotrophy to autotrophy by addition of phosphorus. *Science* 229:1383–6.
- Pinheiro J, Bates D, DebRoy S, R Core Team. 2020. {nlme}: Linear and nonlinear mixed effects models.
- Poole GC. 2002. Fluvial landscape ecology: addressing uniqueness within the river discontinuum. *Freshwater Biology* 47:641–660.
- Raymond PA, Saiers JE, Sobczak WV. 2016. Hydrological and biogeochemical controls on watershed dissolved organic matter transport: pulse-shunt concept. *Ecology* 97:5–16.
- Roberts BJ, Mulholland PJ, Hill WR. 2007. Multiple scales of temporal variability in ecosystem metabolism rates: results from 2 years of continuous monitoring in a forested headwater stream. *Ecosystems* 10:588–606.
- Rocher-Ros G, Sponseller RA, Lidberg W, Mörtz C-M, Giesler R. 2019. Landscape process domains drive patterns of CO₂ evasion from river networks. *Limnology and Oceanography Letters* 4:87–95.
- Roley SS, Tank JL, Griffiths NA, Hall RO, Davis RT. 2014. The influence of floodplain restoration on whole-stream metabolism in an agricultural stream: insights from a 5-year continuous data set. *Freshwater Science* 33:1043–59.
- Saltarelli WA, Dodds WK, Tromboni F, Calijuri M do C, Neres-Lima V, Jordao CE, Palhares JCF, Cunha DGF. 2018. Variation of stream metabolism along a tropical environmental gradient. *Journal of Limnology* 77:359–71.
- Savoy P, Appling AP, Heffernan JB, Stets EG, Read JS, Harvey JW, Bernhardt ES. 2019. Metabolic rhythms in flowing waters: an approach for classifying river productivity regimes. *Limnology and Oceanography* 64:1835–1851.

- Siders AC, Larson DM, Rüegg J, Dodds WK. 2017. Probing whole-stream metabolism: influence of spatial heterogeneity on rate estimates. *Freshwater Biology* 62:711–723.
- Steinman AD, Lamberti GA, Leavitt PR. 2007. Biomass and Pigments of Benthic Algae. In: Hauer FR, Lamberti GA, Eds. *Methods in Stream Ecology*, 2nd edn. Oxford, UK: Academic Press. pp 357–379.
- Summers BM, Horn DJV, González-Pinzón R, Bixby RJ, Grace MR, Sherson LR, Crossey LJ, Stone MC, Parmenter RR, Compton TS, Dahm CN. 2020. Long-term data reveal highly-variable metabolism and transitions in trophic status in a montane stream. *Freshwater Science* 39:241–255.
- Swanson FJ, Kratz TK, Caine N, Woodmansee RG. 1988. Landform effects on ecosystem patterns and processes. *BioScience* 38:92–98.
- Tank JL, Rosi-Marshall EJ, Griffiths NA, Entekin SA, Stephen ML. 2010. A review of allochthonous organic matter dynamics and metabolism in streams. *Journal of the North American Benthological Society* 29:118–146.
- Thorp JH, Thoms MC, DeLong MD. 2006. The riverine ecosystem synthesis: biocomplexity in river networks across space and time. *River Research and Applications* 22:123–147.
- Uehlinger U, Naegeli M, Fisher SG. 2002. A heterotrophic desert stream? The role of sediment stability. *Western North American Naturalist* 62:466–473.
- Ulseth AJ, Bertuzzo E, Singer GA, Schelker J, Battin TJ. 2018. Climate-induced changes in spring snowmelt impact ecosystem metabolism and carbon fluxes in an alpine stream network. *Ecosystems* 21:373–390.
- Van de Bogert MC, Bade DL, Carpenter SR, Cole JJ, Pace ML, Hanson PC, Langman OC. 2012. Spatial heterogeneity strongly affects estimates of ecosystem metabolism in two north temperate lakes. *Limnology and Oceanography* 57:1689–1700.
- Vannote RL, Minshall GW, Cummins KW, Sedell JR, Cushing CE. 1980. River continuum concept. *Canadian Journal of Fisheries and Aquatic Sciences* 37:130–137.
- Venarsky MP, Walters DM, Hall RO, Livers B, Wohl E. 2018. Shifting stream planform state decreases stream productivity yet increases riparian animal production. *Oecologia* 187:167–180.
- Walling DE. 1983. The sediment delivery problem. *Journal of Hydrology* 65:209–237.
- Webster AJ, Cadenasso ML. 2020. Cross-scale controls on the in-stream dynamics of nitrate and turbidity in semiarid agricultural waterway networks. *Journal of Environmental Management* 262:110307.
- Wohl E. 2018. Geomorphic context in rivers. *Progress in Physical Geography: Earth and Environment* 42:841–857.
- Yates AG, Brua RB, Culp JM, Chambers PA. 2013. Multi-scaled drivers of rural prairie stream metabolism along human activity gradients. *Freshwater Biology* 58:675–689.
- Young RG, Huryn AD. 1996. Interannual variation in discharge controls ecosystem metabolism along a grassland river continuum. *Canadian Journal of Fisheries and Aquatic Sciences* 53:2199–2211.
- Zuur A, Ieno EN, Walker N, Saveliev AA, Smith GM. 2009. *Mixed Effects Models and Extensions in Ecology with R*. New York: Springer-Verlag.